

Breast Cancer Trends Among Young Women in the United States

Robert E. Tarone

Background: It has been suggested that exposures associated with industrialization have increased breast cancer risk among young women in recent decades in the United States despite data demonstrating declining breast cancer mortality in birth cohorts born after 1945.

Methods: Trends for in situ and invasive breast cancer incidence rates from 1975 through 2002 among white and black U.S. women ages 20 to 49 are evaluated by decade of age using linear regression analyses.

Results: Despite increasing rates of in situ breast cancer after 1980 reflecting increased use of mammography, invasive breast cancer rates declined for both white and black women under age 50. These declines are consistent with a decrease in birth cohort risk of breast cancer for women born after 1945.

Conclusions: Invasive breast cancer incidence rates are not increasing in young U.S. women despite increases in mammography and trends in known risk factors (eg, reproductive factors) that would predict increasing risk.

It has been reported that breast cancer risk is increasing over time everywhere in the world and that successive birth cohorts of women have had higher levels of risk throughout their lives beginning early in life.¹ This would seem inconsistent with the decreasing risk of breast cancer mortality in young U.S. women due to a decreasing birth cohort risk for women born after 1945.² A possible explanation for this inconsistency is that incidence rates reflect breast cancer risk better than mortality rates.³ An increase in the age-adjusted invasive breast cancer incidence rate for all U.S. women under the age of 50 between years 1975 and 2002 appears to support such an explanation.³

It can be debated whether trends in breast cancer risk (ie, the probability that a woman develops breast cancer) are best represented by trends in incidence rates (which reflect improvements in diagnostic technology as well as risk) or mortality rates (which reflect improvements in cancer detection and cancer treatment as well as risk). Any inference

about breast cancer risk based on the increase between 1975 and 2002 in the age-adjusted breast cancer incidence rate for all U.S. women under age 50, however, must take into account both the marked increase in the use of mammography beginning in the early 1980s and the well-established birth cohort pattern of breast cancer risk.^{4–6} The age-adjusted breast cancer incidence rate in women under the age of 50 would be expected to increase substantially between 1975 and 2002 due to increased use of mammography even if the underlying trend in breast cancer risk is flat or decreasing slowly. In addition, this rate would be expected to increase somewhat between 1975 and the late 1980s due to increasing birth cohort risk for women born before 1945. Accordingly, the observed increase in the age-adjusted incidence rate for women under age 50 between 1975 and 2002 is not sufficient to demonstrate increasing risk of developing breast cancer for women born after 1945.

Secular trends for both in situ and invasive breast cancer incidence rates in young U.S. women by decade of age are evaluated here to determine whether these incidence rates provide evidence of the decreasing birth cohort risk demonstrated previously for U.S. women born after 1945 in the analysis of breast cancer mortality rates.^{5,6}

METHODS

Invasive and in situ breast cancer incidence rates from the Surveillance, Epidemiology, and End Results (SEER) Program of the U.S. National Cancer Institute were calculated using SEER*Stat software.⁷ Directly standardized age-adjusted rates (U.S. 2000 standard population) were calculated separately for white and black women. Regression analyses were performed using annual age-adjusted incidence rates in 3 10-year age groups: 20–29, 30–39, and 40–49 years of age at time of breast cancer diagnosis. The starting years of the regression analyses differed by age group; the recent decrease in risk of dying from breast cancer appears to be a birth cohort phenomenon (ie, decreasing mortality rates for women born after 1945),^{5,6} and so the starting years in the analysis of breast cancer incidence rates for the 2 older decades of age were determined by adding the midpoints of the age intervals to 1945. Thus, the regression analyses were based on rates from 1975 through 2002 for the 20–29 age group, from 1980 through 2002 for the 30–39 age group, and from 1990 through 2002 for the 40–49 age group. Standard 95% confidence intervals (CIs) for the estimated slopes from the regression analyses were calculated using the t-distribution.

Submitted 4 January 2006; accepted 8 March 2006.

From the International Epidemiology Institute, Rockville, MD, and the Department of Medicine and Vanderbilt-Ingram Cancer Center, Vanderbilt University Medical Center, Nashville, TN.

Supplemental material for this article is available with the online version of the journal at www.epidem.com; click on "Article Plus."

Correspondence: Robert Tarone, International Epidemiology Institute, 1455 Research Boulevard, Suite 550, Rockville, MD 20850. E-mail: bob@iei.ws.

Copyright © 2006 by Lippincott Williams & Wilkins

ISSN: 1044-3983/06/1705-0588

DOI: 10.1097/01.ede.0000229195.98786.ee

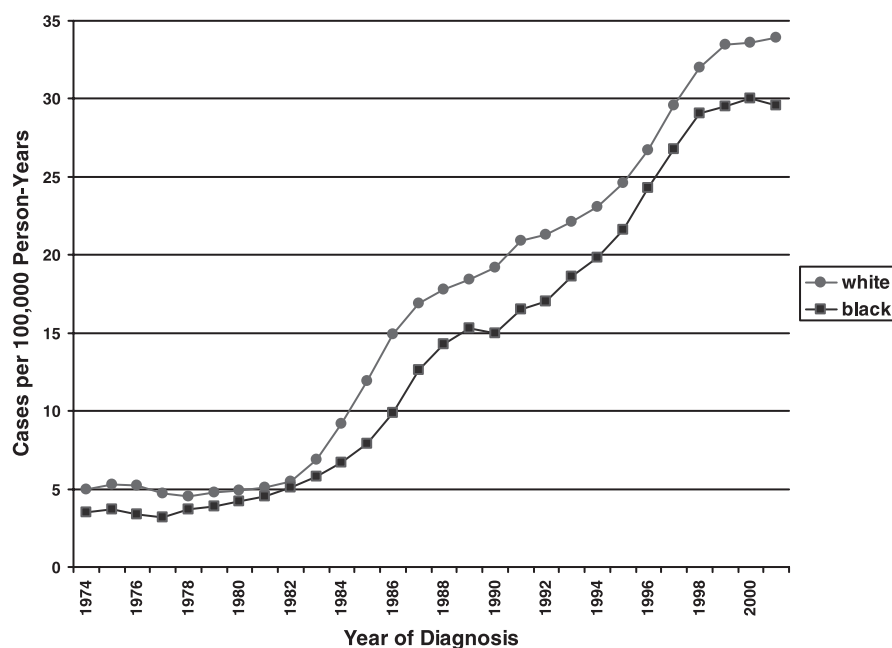


FIGURE 1. Directly standardized age-adjusted in situ breast cancer incidence rates for white and black women in the United States; rates were calculated using SEER*Stat⁷ with the 2000 U.S. population as the standard population. Each point represents the rate for 3 consecutive calendar years, and rates are plotted at the midpoint of the 3-year period.

RESULTS

Secular trends of in situ breast cancer rates provide perhaps the best indicator of increasing mammography use. Figure 1 shows the age-adjusted in situ breast cancer incidence rates for white and black women from 1975 through 2001. To smooth the curves, each point represents the rate for 3 consecutive years, and the rate is plotted at the midpoint of the 3-year period. For both white women and black women, rates increased over 3-fold between 1980 and 1990 and then increased approximately 2-fold between 1990 and 2001. Increases in rates of in situ breast cancer after 1980 were observed for every decade of age beginning with the 20–29 age group (Figs. A1–A6, available with the online version of the article). The marked increase in mammography use reflected by the curves in Figure 1 would also be expected to have a substantial impact on trends in invasive breast cancer incidence rates.

Table 1 shows the results of linear regression analyses of in situ and invasive breast cancer incidence rates by decade of age for white and black women between the ages of 20 and 49. The in situ breast cancer incidence rates increased over time for each decade of age for both white and black women, reflecting the marked increase in utilization of mammography. The SEER invasive breast cancer incidence rates by decade of age in young women have secular trends consistent with the birth cohort pattern of risk observed previously for U.S. breast cancer mortality rates (Figs. A7–A9, available with the online version of the article). Despite the positive bias due to increasing mammography use (ie, bias in terms of incidence rates representing the true trend in underlying breast cancer risk), the invasive breast cancer incidence rates decreased over time for each decade of age for both white and black women (Table 1).

DISCUSSION

Regression analyses of breast cancer incidence rates by decade of age in young women have demonstrated that breast

TABLE 1. Change in Breast Cancer Incidence Rate per Year for White and Black Women by Decade of Age From 20 Through 49 Yr of Age in the United States (SEER Program)

Age Group	In Situ Cancer Slope (95% CI)	Invasive Cancer Slope (95% CI)
20–29*		
White	0.009 (0.004 to 0.015)	–0.022 (–0.046 to 0.003)
Black	0.029 (0.011 to 0.047)	–0.001 (–0.083 to 0.082)
30–39†		
White	0.25 (0.19 to 0.32)	–0.24 (–0.43 to –0.04)
Black	0.15 (0.03 to 0.28)	–0.52 (–0.96 to –0.07)
40–49‡		
White	1.55 (1.10 to 2.00)	–0.54 (–1.32 to 0.24)
Black	1.32 (0.10 to 2.53)	–2.90 (–3.84 to –1.95)

*Linear regression from 1975 through 2002.

†Linear regression from 1980 through 2002.

‡Linear regression from 1990 through 2002.

cancer risk has not been increasing among U.S. women born after 1945. SEER invasive breast cancer incidence rates are consistent with the birth cohort pattern of risk observed previously in analyses of breast cancer mortality rates.^{5,6} Despite marked increases of in situ breast cancer incidence rates (reflecting a substantial increase in the utilization of mammography), invasive breast cancer incidence rates decreased for both white and black women born after 1945. Evidence of a decrease in the birth cohort risk of breast cancer has been noted for women born after World War II in the United States, Canada, and Scotland.^{5,6,8,9} The decreases of invasive breast cancer incidence rates and breast cancer mortality rates among U.S. women born after 1945 have been

observed despite increasing utilization of mammography and despite existing trends in most known or proposed breast cancer risk factors that would predict increasing breast cancer risk (eg, increasing age at first birth, decreasing parity, and both increased and earlier use of oral contraceptives).^{5,10} If, as has been suggested, increasing industrialization also leads to higher breast cancer risk,¹ then the decreasing trends for invasive breast cancer incidence rates and breast cancer mortality rates among young U.S. women are even more surprising. In fact, it follows that unknown and unexplained factors correlated with industrialization must have exerted a profound protective effect against breast cancer. Investigating the paradoxically reduced breast cancer risk among young U.S. women and identifying the unknown protective factor (or factors) should be a major priority in cancer research.

REFERENCES

1. Stevens RG. Circadian disruption and breast cancer: from melatonin to clock genes. *Epidemiology*. 2005;16:254–258.
2. Tarone RE. Circadian disruption and breast cancer [Letter]. *Epidemiology*. 2005;16:710–711.
3. Stevens RG. Circadian disruption and breast cancer [Letter]. *Epidemiology*. 2005;16:711.
4. Chu KC, Tarone RE, Kessler LG, et al. Recent trends in US breast cancer incidence, survival, and mortality rates. *J Natl Cancer Inst*. 1996;88:1571–1579.
5. Tarone RE, Chu KC, Gaudette LA. Birth cohort and calendar period trends in breast cancer mortality in the United States and Canada. *J Natl Cancer Inst*. 1997;89:251–256.
6. Chu KC, Tarone RE, Brawley OW. Breast cancer trends of black women compared with white women. *Arch Fam Med*. 1999;8:521–528.
7. National Cancer Institute. *SEER*Stat Software*, version 6.1.4. Bethesda, MD: Division of Cancer Control and Population Sciences, Surveillance Research Program, Cancer Statistics Branch; released April 2005.
8. Boyle P, Robertson C. Breast cancer and colon cancer incidence in females in Scotland, 1960–1984. *J Natl Cancer Inst*. 1987;79:1175–1179.
9. Robertson C, Boyle P. Age-period-cohort analysis of chronic disease rates. I: modeling approach. *Stat Med*. 1998;17:1305–1323.
10. Tarone RE, Chu KC. Response. Re: Recent trends in US breast cancer incidence, survival and mortality rates. *J Natl Cancer Inst*. 1997;89:1811–1812.